

Global CO₂ Emissions

Trends, Regional Shifts, and Per Capita Inequalities

Scope: 1750–2024 | Data: Global Carbon Budget 2025, NASA/NOAA

Prepared for policy analysts and climate negotiators

Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

429

parts per million (Feb 2026)

NOAA Mauna Loa Observatory

Pre-industrial baseline: ~280 ppm

Current level is >50% above baseline

Equivalent to 1.5× the natural increase at the end of the last ice age — achieved in ~200 years vs. thousands of years

Driven primarily by burning coal, oil, and natural gas

280 ppm (1750)

~315 ppm (1958, Keeling Curve begins)

429 ppm (2026)

Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Bt to 35+ Bt

6 Bt

1950

Post-war baseline

20 Bt

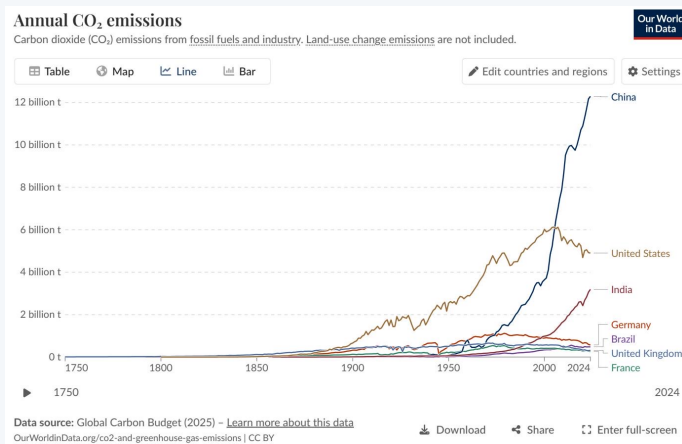
1990

Quadrupled in 40 years

35+ Bt

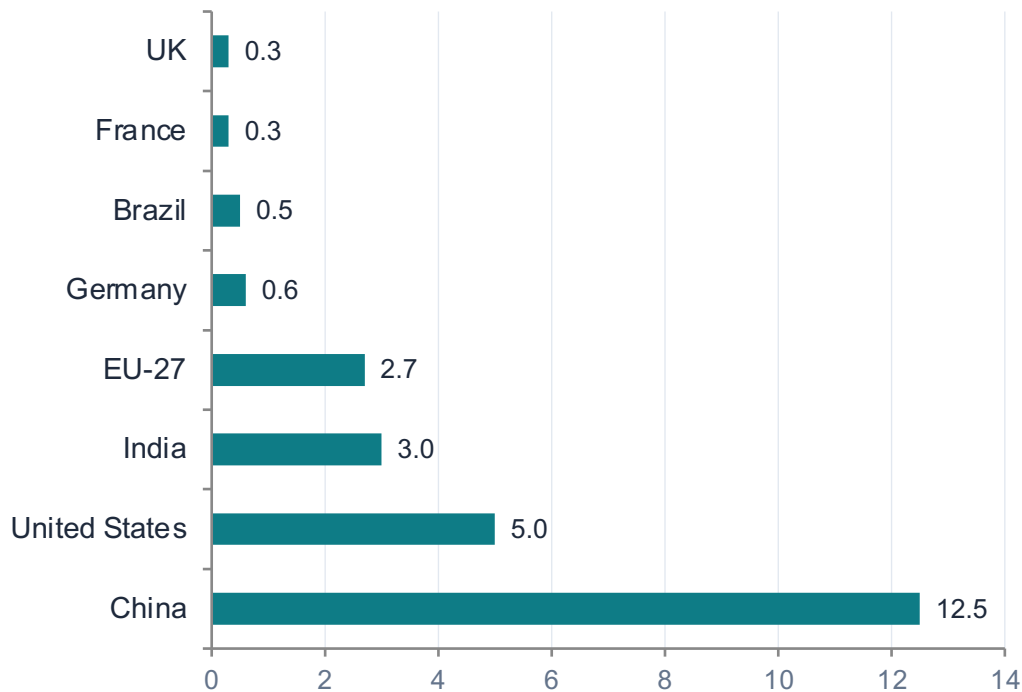
2024

Still rising, not yet peaked



Source: Global Carbon Budget 2025 (figure_3)

China Now Emits Over 12 Billion Tonnes — More Than Double the US



China: ~27% of global total

US peaked at ~6 Bt around 2005; now declining

India ~8% of total and rising fast

EU-27 ~7%, declining steadily

Reference: figure_3.jpg

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

1900



The Industrial Revolution was concentrated in Europe and North America

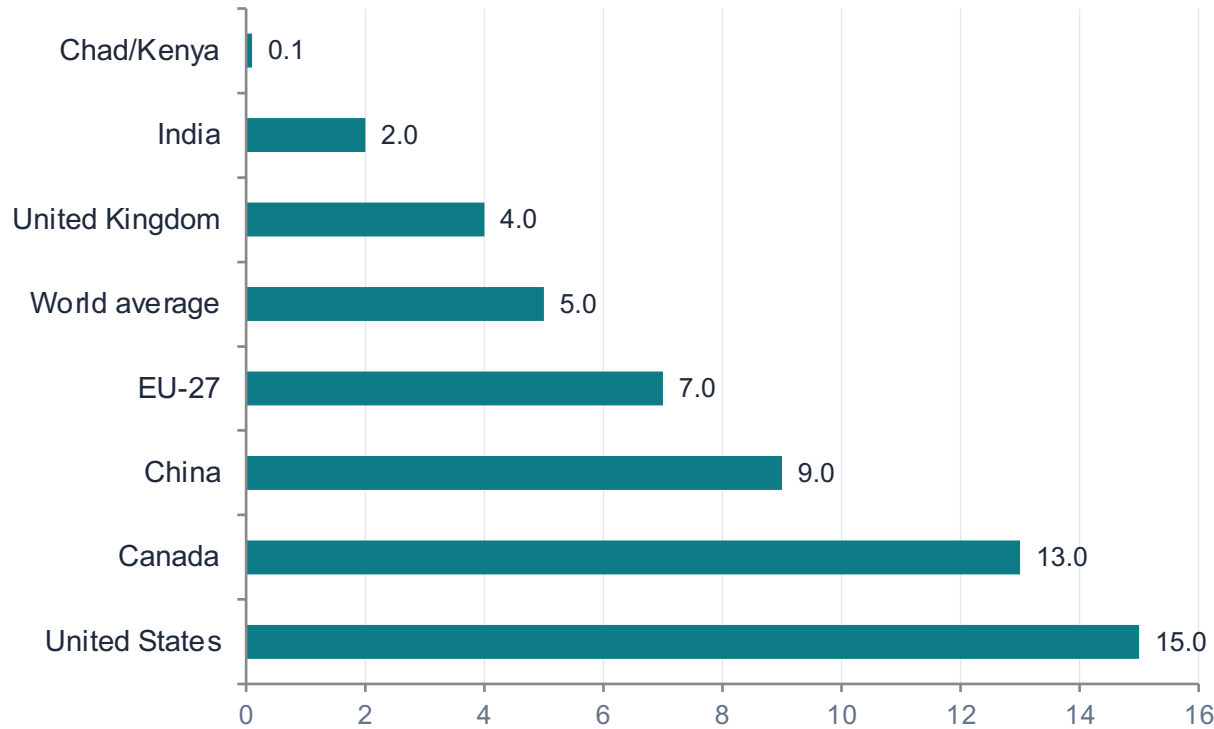
2024



Asia is home to ~60% of world population; Africa & S. America combined emit ~6-8%

A dramatic geographic rebalancing in just over a century

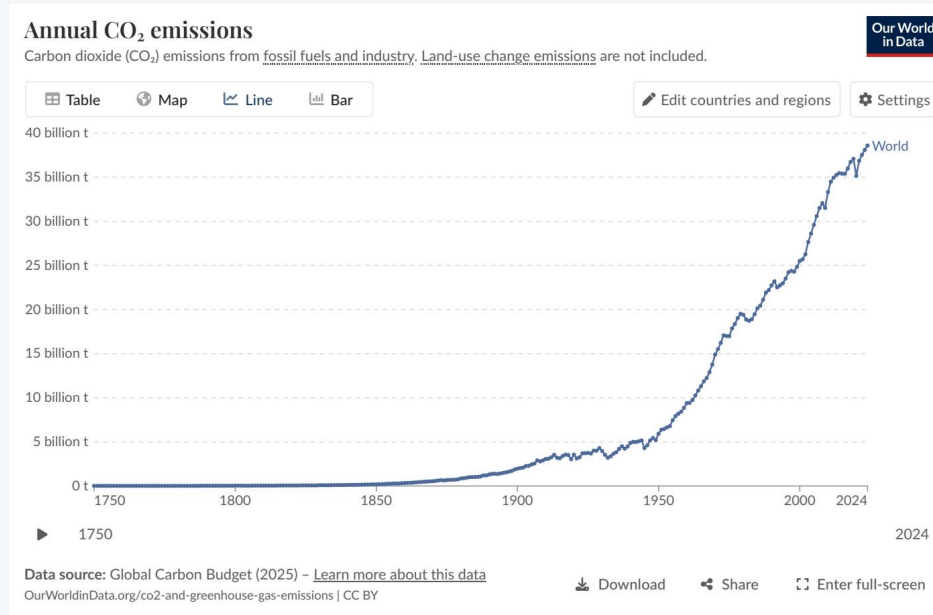
Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



150×

gap between highest and lowest per capita emitters

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase



Declining (blue): US, most of Europe (−1% to −5%)

Increasing (orange/red): Russia, SE Asia, parts of Africa (+2% to +10%)

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar prosperity levels can have very different per capita emissions depending on their energy sources.

	France / UK	Germany / Netherlands
Primary Energy Sources	High nuclear + renewables	Higher fossil fuel share (coal, gas)
Per Capita CO ₂	~4 tonnes/year	~8–9 tonnes/year
Living Standards	High (comparable GDP/capita)	High (comparable GDP/capita)
Key Difference	Nuclear baseload + wind/solar growth	Coal phase-out still incomplete; natural gas dependence

Policy and energy mix choices — not just prosperity — determine per capita outcomes

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

- Global emissions exceed 35 Bt/year and have not yet peaked, despite slowing growth rates.
- China alone accounts for >25% of global emissions (~12.5 Bt), more than double the US (~5 Bt); India (~3 Bt) is third and rising fast.
- The geographic center has shifted dramatically: from >90% in Europe/US in 1900 to Asia producing ~50% today.
- Per capita inequalities remain extreme — a 150× gap between highest (US, Australia ~15 t) and lowest (Chad ~0.1 t) emitters.
- Energy mix and policy choices matter: countries with similar prosperity (France vs. Germany) can have very different per capita emissions.